

Accelerometer-measured physical activity, sedentary behavior, and incident neuropsychiatric diseases: a large prospective cohort study of 73411 participants

Wu JY, Li YZ, Zhang DD, Gao PY, Fu Y, Zhang W, Feng JF, Zhang YR, Cheng W, Yu JT. *Molecular Psychiatry*. 2026 Mar; 31(3):1485-1499 (published online 20 Sep 2025). DOI: 10.1038/s41380-025-03268-7 PMID: 40975752

Abstract

Physical activity and sedentary behavior are closely associated with neuropsychiatric diseases, while previous studies have mainly relied on self-reported data, which have been shown to be inconsistent with objectively measured metrics. Using data from 73,411 participants, objectively quantifying physical activity through energy expenditure and time allocation, we identified longitudinal associations between physical activity, sedentary behavior, and neuropsychiatric diseases using Cox proportional hazards models. Our findings indicated that moderate to vigorous physical activity energy expenditure showed the strongest protective effect on neuropsychiatric diseases (hazard ratios: 0.60-0.86, all FDR-Q < 0.001), while an increased proportion of sedentary time was identified as a risk factor for neuropsychiatric diseases (HRs: 1.05-1.54, FDR-Q < 0.05). Restricted cubic spline analyses demonstrated an L-shaped association linking physical activity and neuropsychiatric diseases and an inverted L-shaped relationship between sedentary behavior and dementia. Linear regression models further linked physical activity and sedentary behavior to cognitive function, physical function, and mental health. Key brain regions related to these behaviors included the lateral occipital cortex, cuneus, pallidum, and accumbens. Proteomics and metabolic analyses identified significant involvement of ITGAV protein and high-intensity lipoprotein. Structural equation modeling elucidated that inflammation and metabolism mediate the relationship between physical activity, sedentary behavior, and neuropsychiatric diseases. Overall, our study provides critical evidence of the link between physical activity, sedentary behavior, and neuropsychiatric diseases, shedding light on potential neurobiological mechanisms underlying these associations.

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